

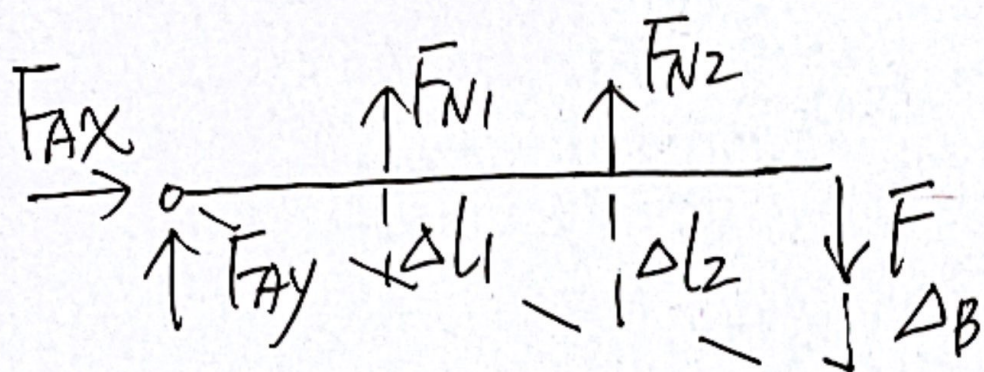
- DBCAA BCD BC

二. 1. 受力图

2'+2'

$$\sum M_A(\bar{F}) = 0$$

$$F_{N1} \cdot 1 + F_{N2} \cdot 2 - F \cdot 3 = 0 \quad (1)$$



比例:

$$\Delta l_1 = \frac{1}{2} \Delta l_2 = \frac{1}{3} \Delta B \quad 2'$$

物理 $\frac{F_{N1} \cdot 1}{EA} = \frac{F_{N2} \cdot 1}{EA} \cdot \frac{1}{2} \Rightarrow F_{N1} = \frac{1}{2} F_{N2} \quad 2' \quad (2)$

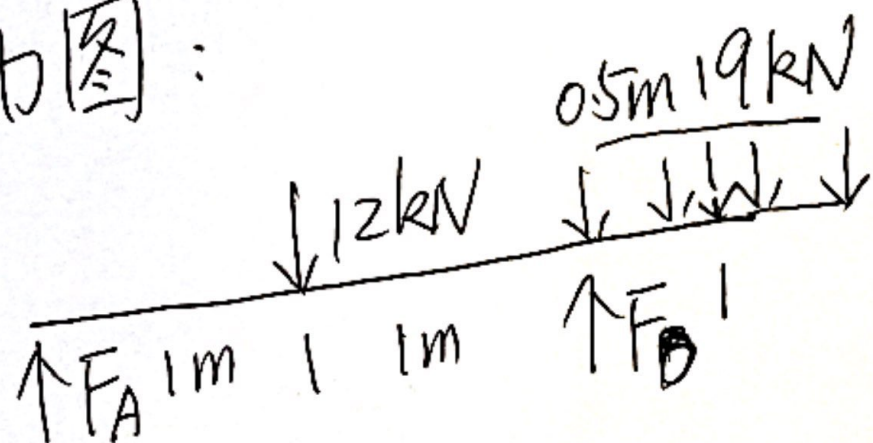
(1) (2) $F_{N1} = \frac{3}{5} F = 60 \text{ kN}$ $F_{N2} = \frac{6}{5} F = 120 \text{ kN} \quad 4'$

$$\sigma_1 = \frac{F_{N1}}{A} = \frac{60 \times 10^3 \text{ N}}{1000} = 60 \text{ MPa} \quad \sigma_2 = 120 \text{ MPa}$$

B点位移: $\Delta B = 3 \Delta l_1 = 3 \frac{\sigma_1}{E} \cdot l$

$$= 3 \times \frac{60 \text{ MPa}}{200 \times 10^3 \text{ MPa}} \times 1 \text{ m} = 3 \times 0.3 = 0.9 \text{ mm} \quad 2' \quad (\downarrow)$$

2. 受力图:

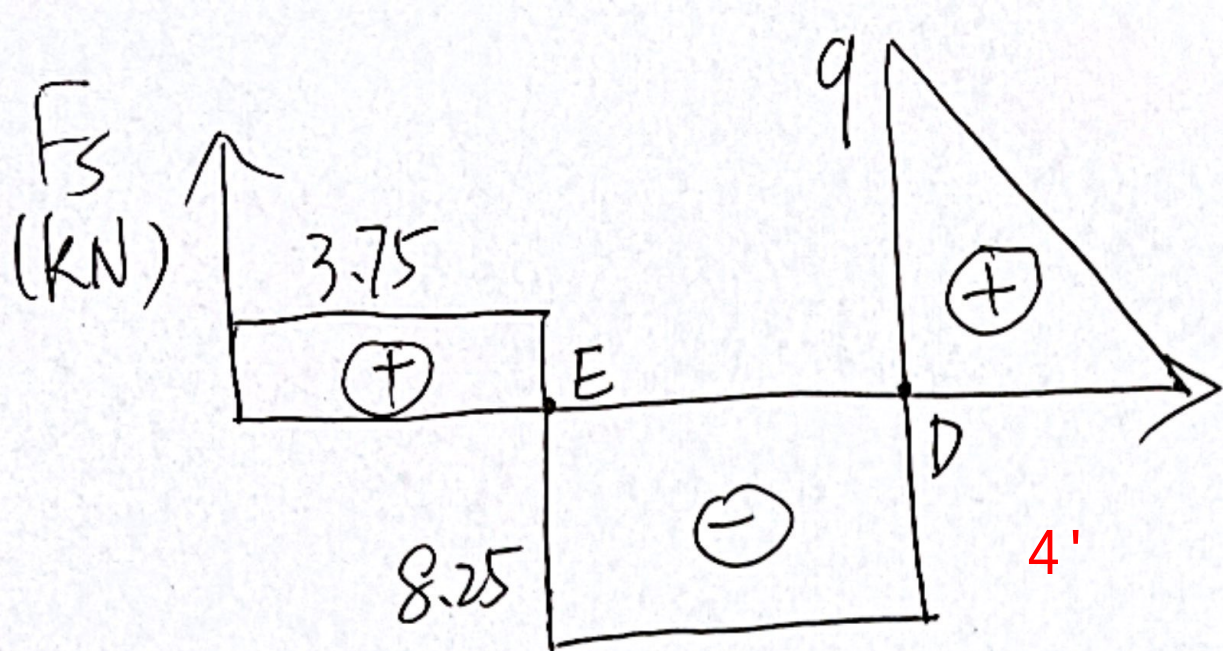


$$\sum M_A(F) = 0 \quad 12 \times 1 + 9 \times 2.5 - F_B \times 2 = 0$$

$$F_B = 17.25 \text{ kN} \uparrow$$

$$\sum F_y = 0$$

$$F_A = 12 + 9 - 17.25 = 3.75 \text{ kN} \uparrow \quad 2'$$

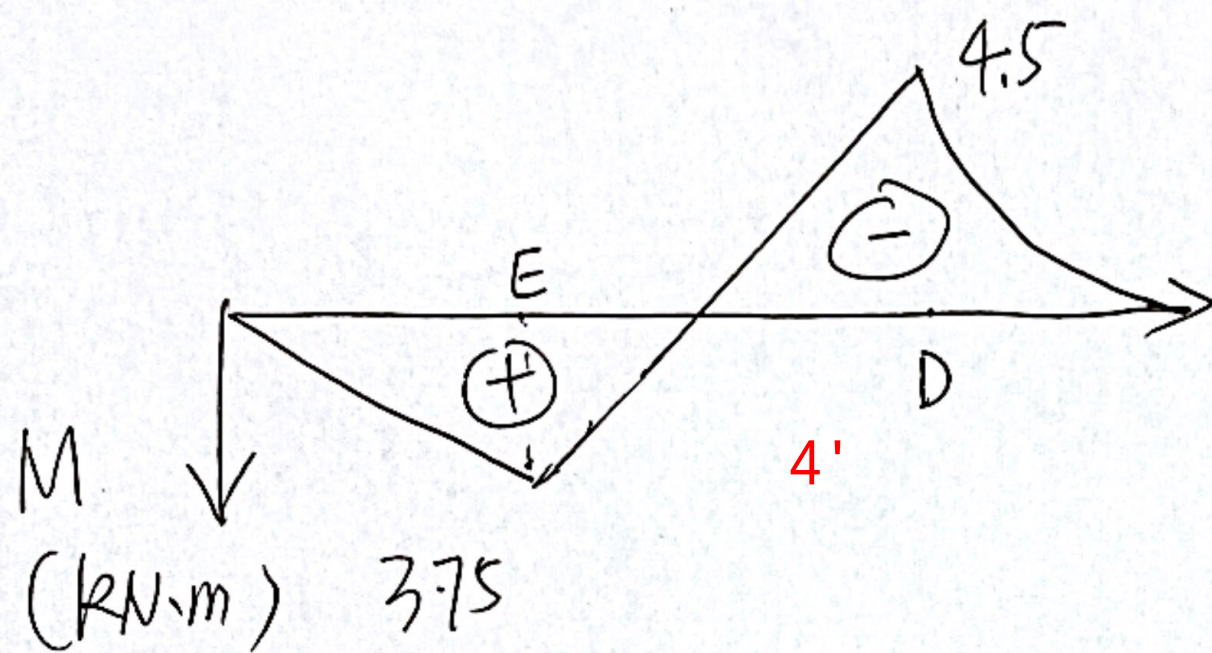


正应力强度

$$\text{拉: } \sigma_{ET} = \frac{M_E \cdot y_1}{I_z}$$

$$= \frac{3.75 \times 10^6 \times 140}{40 \times 10^6}$$

$$= 13.125 \text{ MPa} < [\sigma_t]$$



$$\sigma_{DT} = \frac{4.5 \times 10^6 \times 60}{40 \times 10^6}$$

$$= 6.75 \text{ MPa} < [\sigma_t]$$

$$\text{压: } \sigma_{DT} = \frac{4.5 \times 10^6 \times 140}{40 \times 10^6}$$

$$= 15.75 \text{ MPa} < [\sigma_c]$$

$$\sigma_{EL} = 5.625 \text{ MPa}$$

故满足强度要求。 1'

3. $\sigma_x = 80 \text{ MPa}$ $\tau_x = -30 \text{ MPa}$

$$\sigma_y = 0 \text{ MPa}$$

$$\sigma_1 = 40 + 50 = 90 \text{ MPa}$$

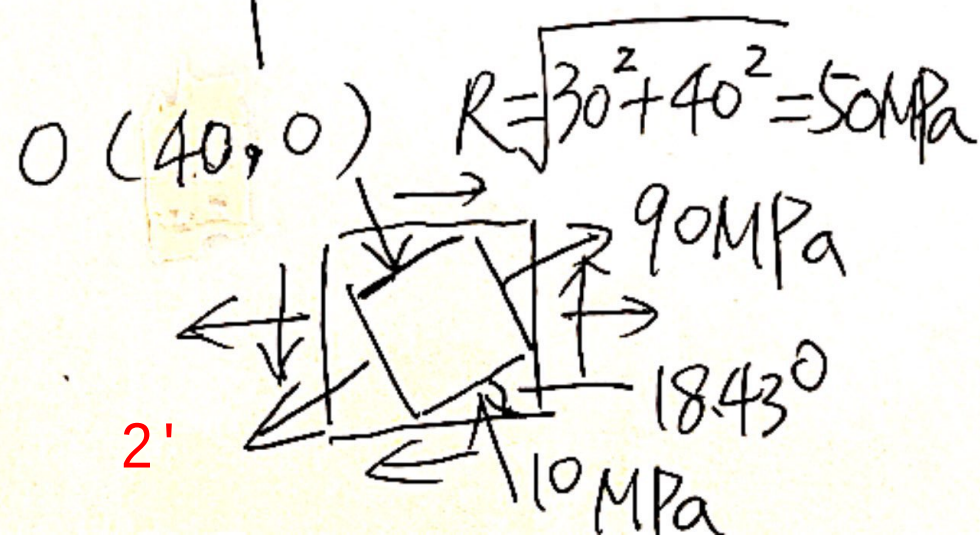
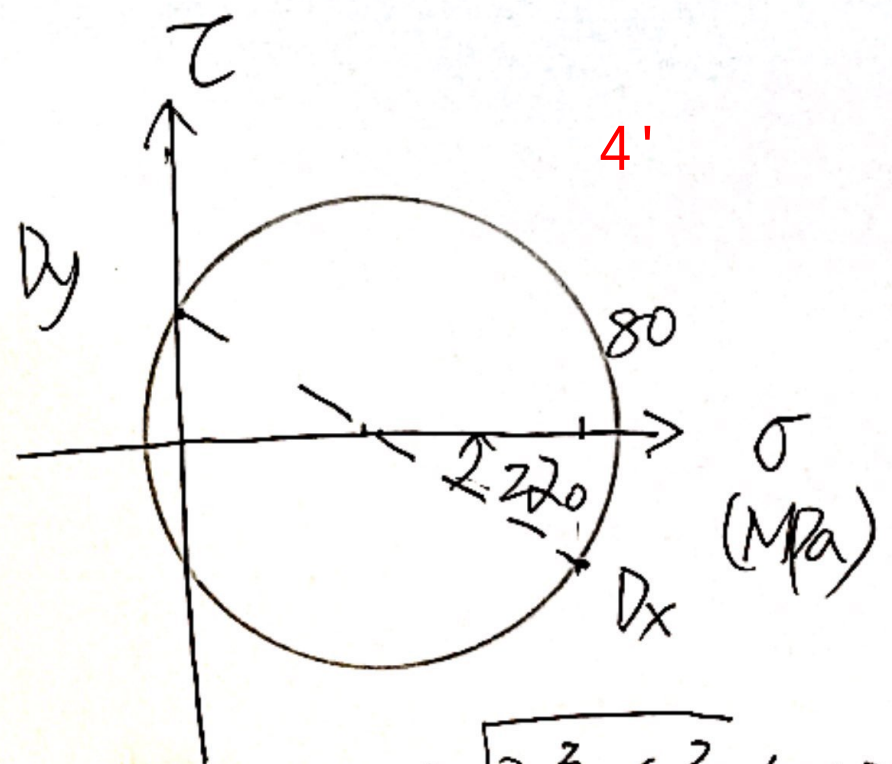
$$\sigma_2 = 0$$

$$\sigma_3 = 40 - 50 = -10 \text{ MPa}$$

$$\tan 2\alpha_0 = \frac{30}{40} \quad 2\alpha_0 = 18.43^\circ$$

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = 50 \text{ MPa}$$

$$\sigma_{13} = \sigma_1 - \sigma_3 = 100 \text{ MPa}$$

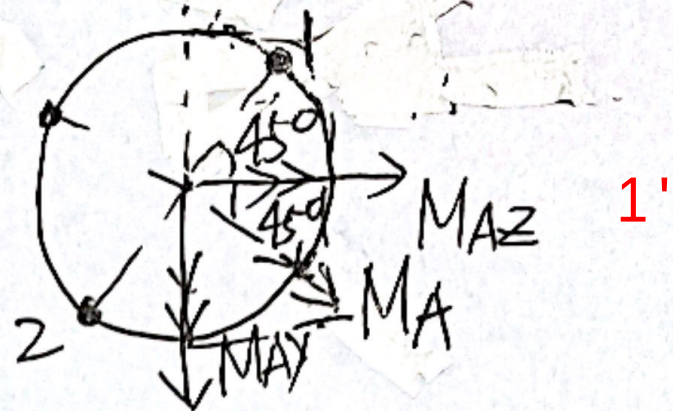


4. 危险截面: A

$$T = Fa = 900 \text{ N} \cdot \text{m} \text{ (指向截面)}$$

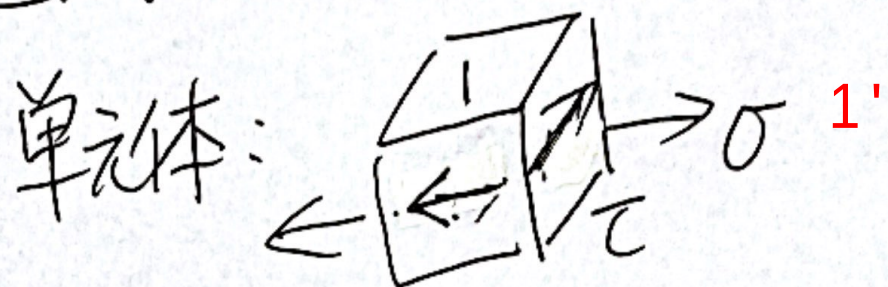
$$M_{Az} = 2Fa = 1800 \text{ N} \cdot \text{m} \quad 3'$$

$$M_{Ay} = \frac{2}{3}F \cdot 3a = 1800 \text{ N} \cdot \text{m}$$



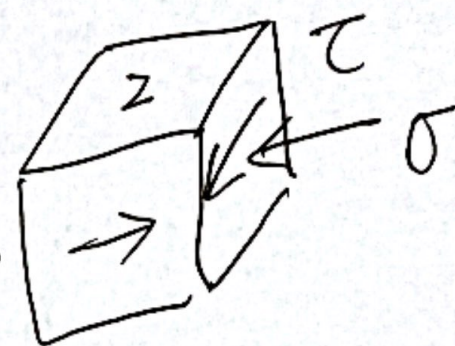
危险点: 1, 2

$$\sigma_{r4} = \frac{\sqrt{M_{Ay}^2 + M_{Az}^2 + 0.75T^2}}{\frac{\pi}{32}d^3} \leq [\sigma] \quad 3'$$



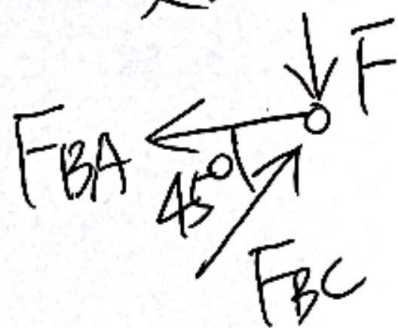
即

$$\frac{\sqrt{900^2 \times 0.75 + 1800^2 + 1800^2} \times 10^3}{\frac{\pi}{32}d^3} \leq 160 \rightarrow \quad 1'$$



$$d^3 \geq 169483 \text{ mm}^3 \quad d \geq 55.34 \text{ mm} \quad \text{取 } d = 56 \text{ mm} \quad 1'$$

5. 受力图



$$\sum F_y = 0 \quad F_{BC} \cos 45^\circ - F = 0 \quad F_{BC} = \sqrt{2}F \quad 2'$$

BC杆: $i = \frac{d}{4} = 11.25 \text{ mm} \quad \mu = 1$

$$\lambda = \frac{\mu l}{i} = \frac{1 \times 1250}{11.25} = 111.1 > \lambda_p \text{ 细长杆} \quad 2'$$

$$\lambda_p = \sqrt{\frac{\pi^2 E}{\sigma_p}} = \sqrt{\frac{\pi^2 \times 200 \times 10^3}{200}} = 99.3$$

$$F_{cr} = \frac{\pi^2 E}{\lambda^2} \cdot \frac{\pi d^2}{4} = 254.29 \text{ kN} \quad 4$$

$$\sqrt{2}F \leq \frac{F_{cr}}{4}$$

$$F \leq \frac{254.29}{4\sqrt{2}} = 44.95 \text{ kN} \quad 2$$